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5G to make India smarter: How far is the reality?

of \$1 trillion, TRAI said. The superfast 5G network and associated technologies will help drive digital transformation in the country and achieve the goal of digital empowerment. The telecom sector in India currently reeling under huge debts, is banking heavily on 5G to increase their average revenue per user (ARPU) for growth. As per GSMA Intelligence, 5G connection in India is expected to reach 6% of the total population by 2025 (72 million) and 93% by 2040.

The momentum has already begun in India with several smartphone makers launching 5G-enabled devices even before the official launch of 5G. At the end of 2020, India had nearly 2 million 5G-enabled devices (Nokia MBiT Index), and 5G smartphone shipments in India is expected to hit the 38 million units mark in India in 2021 – 21% share of the total smartphone market in the same year (Counterpoint Research).

The immediate beneficiaries of 5G in India will be smartphone users. 5G offers much faster connectivity with up to 20 Gbps peak data rates and 100+ Mbps average data rate, as well as an average latency of below 10 milliseconds and less than one millisecond in best-case scenarios. Thus, 5G momentum builds on the expectations of delivering enhanced Mobile Broadband (eMBB) which will spur the usage of mobile video, OTT applications, AR/VR and cloud gaming.

In India, 5G can also support several use cases like telemedicine, rural education, crop monitoring, energy management, and AR/VR applications across a wide range of consumer segments. In agriculture, 5G can address some of the critical challenges like water scarcity faced by farmers across the globe. They can leverage connected technologies, data analytics, smart sensors powered by 5G to monitor the field

to ensure optimal use of the resources while maximizing agricultural outputs.

Nokia is contributing to the 5G evolution through innovative solutions that are also futureproof. The Nokia Smart Factory in India is a significant step forward, as it has emerged as the first Indian factory to start 5G New Radio production (2018), making in India for the world. We are committed to contributing towards the Make in India vision and drive a self-sustainable 5G ecosystem in the country; our R&D initiatives are designed along this line.

While 5G can complement 4G networks in most areas, the viability of 5G as a rural broadband solution is yet to be materialized. The development around edge computing proves vital in this scenario as it can facilitate network access near the premise, thereby addressing the latency challenges in these applications. It is also important that all the stakeholders, including the government, CSPs, telecom vendors and businesses make proactive steps to make it a reality. The government needs to provide a framework that encourages investment in 5G while CSPs and the industry as a whole need to educate the consumers and enterprises on the benefits of 5G and how it can transform their lives and business.

Looking ahead to 2030, Nokia will redouble its efforts to act responsibly, ethically and sustainably, through a four-pronged strategy: promote equality by making 5G accessible to all, building digital trust by giving more control over personal data, put people first by making life better with 5G, and ensure sustainability by promoting greener initiatives. As 5G deployments progress globally, the time is also right to build a vision of the future into the research of 6G. Nokia is taking the lead in building the foundation and roadmap for 6G. 

On July 31, 1995, the first wireless call in India was made between erstwhile Union Telecom Minister Sukh Ram and Chief Minister of West Bengal Jyoti Basu on Nokia mobile phones on a GSM network operated by Modi-Telstra. Now, 25 years hence, as India awaits the fifth generation of telecommunications (5G), the horizon looks entirely different.

Globally, the rate of 5G uptake has been faster compared to the previous generation of mobile technologies. It is estimated that 5G grows nearly 3x faster than 4G, and with this rate, it can acquire 1 billion global users in just three-and-a-half years. Research conducted by Nokia Bell Labs in 2020 found that 5G has the potential to contribute \$8 trillion to global GDP by 2030, with industries as diverse as mining, entertainment, transportation, and healthcare all set to reap enormous benefits. It has the power to transform the world's education and employment. Globally around 1.4 billion people are using mobile devices to educate themselves or their children. It is estimated that 5G has the potential to meet the needs of the 262 million children and youth who are not attending any kind of school.

In India alone, 5G has the potential to create a cumulative economic impact